

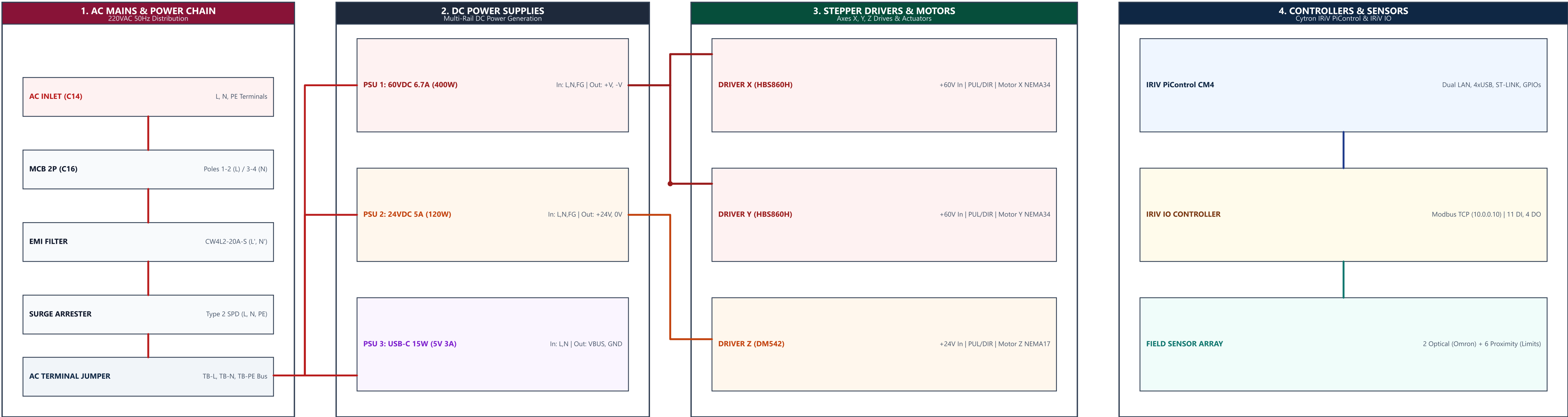
PROJECT: NARIT SMART VENDING MACHINE | SYSTEM OVERVIEW & PIN-TO-PIN ARCHITECTURE

SECTION: OVERVIEW | IEC 60204-1 / ISO 7200 STANDARDS

PIN-TO-PIN WIRING METHODOLOGY & SPECIFICATION:

This document explicitly defines the point-to-point electrical connections between terminal blocks, power buses, controllers, motor drives, and sensors. Every terminal pin, wire gauge, and color code is rigorously specified.

Standard: IEC 60204-1
Voltage: 220VAC / 60V / 24V / 5V
Rated Current: 16A Max

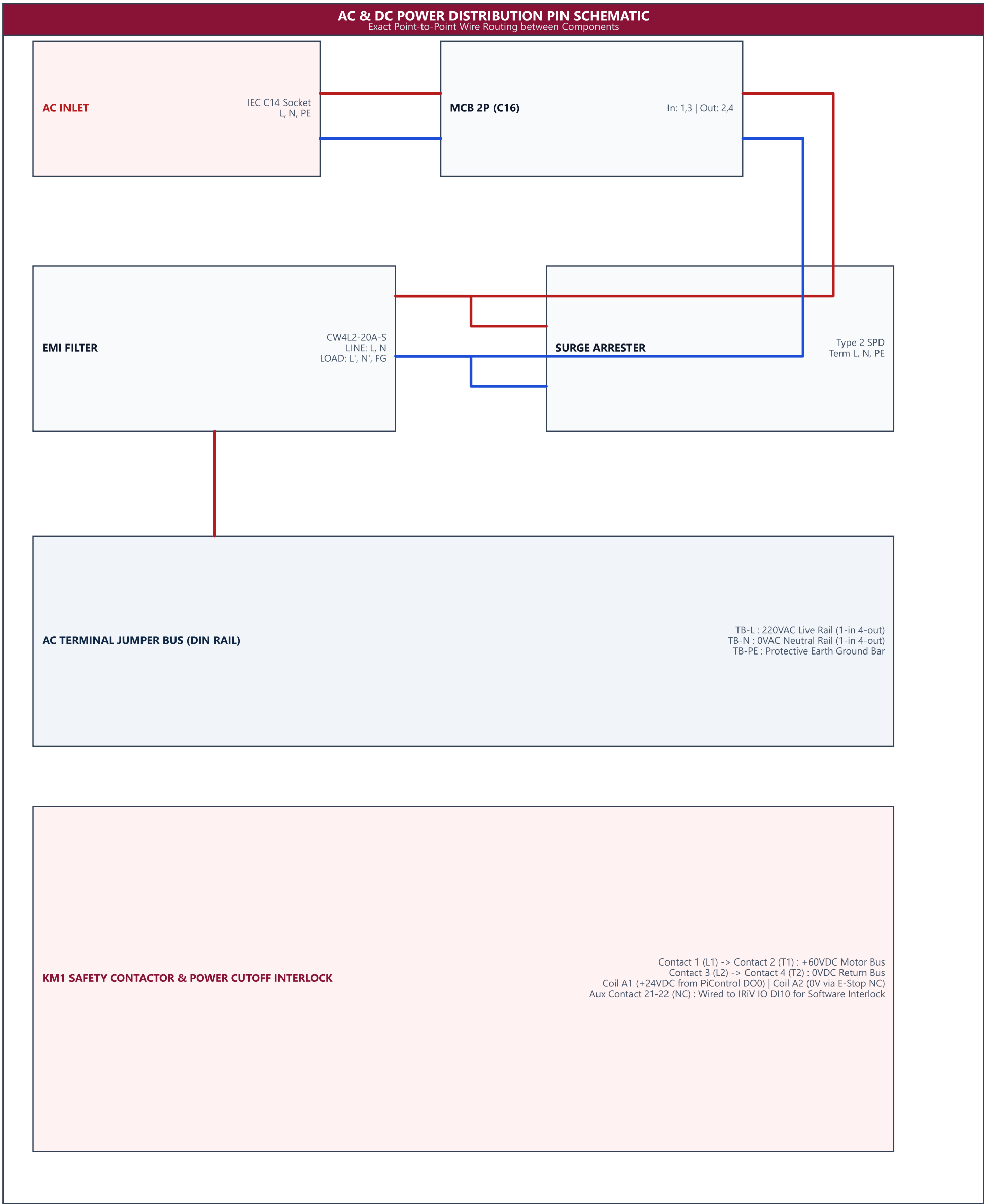


MASTER PIN-TO-PIN CONNECTION SCHEDULE
Explicit Terminal-to-Terminal Mapping across all Cabinets and Rails

WIRE	ORIGIN COMPONENT	ORIGIN PIN	TARGET COMPONENT	TARGET PIN	WIRE SPEC / COLOR	ELECTRICAL FUNCTION
W-01	AC Mains Inlet C14	Pin L (Live)	MCB 2P 16A	Pole 1 (L_in)	2.5 mm² Brown	AC 220V Input to Breaker
W-02	MCB 2P 16A	Pole 2 (L_out)	EMI Noise Filter	Terminal LINE (L)	2.5 mm² Brown	Protected AC Live to Filter
W-03	EMI Noise Filter	Terminal LOAD (L')	AC Terminal Bus	Terminal TB-L	2.5 mm² Brown	Filtered AC Distribution Bus
W-04	AC Terminal Bus	Terminal TB-L	PSU 1 (60V 6.7A)	Terminal L (AC)	2.5 mm² Brown	Feeds 60V Motor Power Supply
W-05	PSU 1 (60V 6.7A)	Terminal +V (DC)	KM1 Safety Contactor	Main Contact 1 (L1)	2.5 mm² Red	60VDC Motor Bus to Contactor
W-06	KM1 Safety Contactor	Main Contact 2 (T1)	Drivers X & Y	Terminal AC / V+	2.5 mm² Red	Switched 60VDC to X/Y Drives
W-07	PSU 2 (24V 5A)	Terminal +24V	IRIV IO / Sensors	Terminal 24V / Anode	1.5 mm² Orange	+24VDC Control & V-PULSE Bus
W-08	PSU 3 (5V 3A)	USB-C VBUS Pin	IRIV PiControl CM4	USB-C Power Port	18 AWG Shielded	+5.1VDC Isolated CM4 Supply
W-09	IRIV PiControl CM4	RJ45 Port (eth1)	IRIV IO Controller	RJ45 Modbus Port	Cat6 UTP (Blue)	Modbus TCP Industrial LAN (10.0.0.2 -> 10.0.0.10)
W-10	STM32 NUCLEO-G491RE	Morpho CN10 pin 23	Driver X (HBS860H)	Terminal PUL- (via Q1)	0.5 mm² Blue (Pair)	Hardware Timer Pulse (TIM1_CH1 / PA8)
W-11	Sensor Array (Limits)	Black Signal Wire	IRIV IO Controller	Terminals DI0 - DI5	0.5 mm² Shielded	24V Normally Closed Limit Triggers
W-12	KM1 Aux Contact	Terminal 21-22 (NC)	IRIV IO Controller	Terminal DI10	0.5 mm² Shielded	Fail-Safe E-Stop & Contactor Feedback

PROJECT: NARIT SMART VENDING MACHINE | SECTION 1: AC MAINS & POWER SUPPLIES PIN-TO-PIN WIRING

SECTION: SECTION 1 | IEC 60204-1 / ISO 7200 STANDARDS



PIN-TO-PIN CONNECTION SCHEDULE (SECTION 1)						
Origin Terminal -> Target Terminal, Wire Gauge, Color, and Voltage						
NO	FROM COMPONENT	PIN / TERM	TO COMPONENT	PIN / TERM	WIRE COLOR / GAUGE	SIGNAL / VOLT
1	AC Inlet C14	Line (L)	MCB 2P 16A	Pole 1 (L_in)	Brown 2.5 mm²	220VAC Phase
2	AC Inlet C14	Neut (N)	MCB 2P 16A	Pole 3 (N_in)	Blue 2.5 mm²	0VAC Neutral
3	AC Inlet C14	Earth (PE)	AC Terminal Bus	TB-PE Bar	Grn/Yel 2.5 mm²	Chassis Earth
4	MCB 2P 16A	Pole 2 (L_out)	EMI Filter	Terminal L	Brown 2.5 mm²	220VAC Line
5	MCB 2P 16A	Pole 4 (N_out)	EMI Filter	Terminal N	Blue 2.5 mm²	0VAC Neutral
6	EMI Filter	Terminal L'	AC Terminal Bus	TB-L Bus	Brown 2.5 mm²	Filtered 220V
7	EMI Filter	Terminal N'	AC Terminal Bus	TB-N Bus	Blue 2.5 mm²	Filtered 0V
8	EMI Filter	Terminal FG	AC Terminal Bus	TB-PE Bar	Grn/Yel 2.5 mm²	Earth Ground
9	AC Terminal Bus	TB-L Bus	Surge Arrester	Terminal L	Brown 2.5 mm²	220VAC Surge
10	AC Terminal Bus	TB-N Bus	Surge Arrester	Terminal N	Blue 2.5 mm²	0VAC Surge
11	Surge Arrester	Terminal PE	AC Terminal Bus	TB-PE Bar	Grn/Yel 2.5 mm²	Earth Discharge
12	AC Terminal Bus	TB-L Bus	PSU 1 (60V)	Terminal L	Brown 2.5 mm²	220VAC Feed
13	AC Terminal Bus	TB-N Bus	PSU 1 (60V)	Terminal N	Blue 2.5 mm²	0VAC Return
14	AC Terminal Bus	TB-L Bus	PSU 2 (24V)	Terminal L	Brown 1.5 mm²	220VAC Feed
15	AC Terminal Bus	TB-N Bus	PSU 2 (24V)	Terminal N	Blue 1.5 mm²	0VAC Return
16	AC Terminal Bus	TB-L Bus	PSU 3 (5V)	Plug Live	Brown 1.5 mm²	220VAC Feed
17	AC Terminal Bus	TB-N Bus	PSU 3 (5V)	Plug Neut	Blue 1.5 mm²	0VAC Return
18	PSU 1 (60V)	Terminal +V	KM1 Contactor	Contact 1 (L1)	Red 2.5 mm²	+60VDC Motor
19	PSU 1 (60V)	Terminal -V	KM1 Contactor	Contact 3 (L2)	Black 2.5 mm²	0VDC Return
20	KM1 Contactor	Contact 2 (T1)	Drivers X & Y	Terminal AC/V+	Red 2.5 mm²	+60VDC Switched
21	KM1 Contactor	Contact 4 (T2)	Drivers X & Y	Terminal AC/V-	Black 2.5 mm²	0VDC Switched
22	PiControl CM4	Terminal DO0	KM1 Contactor	Coil A1 (+24V)	Orange 1.0 mm²	Coil Drive / Reset
23	KM1 Contactor	Coil A2 (0V)	E-Stop Button	NC Contact 1	Red 1.0 mm²	Series E-Stop Loop
24	KM1 Contactor	Aux Contact 21-22	IRiV IO Modbus	Terminal DI10	Shielded 0.5 mm²	Contactor Feedback

PROJECT: NARIT SMART VENDING MACHINE | SECTION 2: CONTROLLERS & COMMUNICATION PIN-TO-PIN WIRING

SECTION: SECTION 2 | IEC 60204-1 / ISO 7200 STANDARDS

CONTROLLER TERMINALS & PORT MAPPING

PiControl CM4 & IRiV IO Physical Ports and Terminals

Cytron IRiV PiControl CM4 (MAIN CPU)

eth0 : 192.168.70.80 (Flask Web Kiosk UI Port 80, REST API, MQTT)
eth1 : 10.0.0.2 (OT Industrial Modbus Master to IRiV IO)
USB 1 : 2D QR Scanner | USB 2 : FHD Web Camera
USB 3 : Audio Speaker (3W) | USB 4 : ST-LINK V3 VCP (115200 8-N-1)
Power : USB-C (5V 3A) + 24V Aux Terminal Block

PiControl LOCAL ISOLATED GPIO TERMINAL BLOCK

DI0 (GPIO 13) : X_DRIVE_ALM (From Driver X ALM- Opto)
DI1 (GPIO 17) : Y_DRIVE_ALM (From Driver Y ALM- Opto)
DI2 (GPIO 27) : X_PEND (From Driver X PEND- Opto)
DI3 (GPIO 22) : Y_PEND (From Driver Y PEND- Opto)
DO0 (GPIO 23) : XY_DRIVE_POWER_KM1 (Safety Contactor Driver)
24V IN / 0V IN : Power terminals from PSU 2 (+24V / 0V Rail)

Cytron IRiV IO (MODBUS TCP SLAVE @ 10.0.0.10:502)

RJ45 Port : Modbus TCP from PiControl eth1 (Unit ID 255)
Power Terminals : +24V, 0V, S/S (S/S jumpered to 0V for PNP mode)
DI0 - DI5 : Limits Min/Max X, Y, Z (Normally Closed Proximity)
DI6 - DI7 : Z Home Sensor (NO) & Product Drop Alignment Parking
DI8 - DI9 : Completed Drop (Omron Beam) & Pickup Door Flap Access
DI10 : KM1 Safety Contactor Feedback (Fail-Safe NC Loop)
DO0 - DO3 : Ready, Moving, Alarm Indicators & Dispense Solenoid

PIN-TO-PIN CONNECTION SCHEDULE (SECTION 2)

Controller Pinout, Communication Lines, and Field I/O Termination

NO	FROM COMPONENT	PIN / TERM	TO COMPONENT	PIN / TERM	WIRE COLOR / GAUGE	SIGNAL / FUNCTION
1	PiControl CM4	RJ45 eth1	IRiV IO Controller	RJ45 Modbus	Cat6 UTP Blue	Modbus TCP (10.0.0.2/10)
2	PiControl CM4	USB 4 Port	NUCLEO-G491RE	ST-LINK VCP	Shielded USB-A	Safe-Link v3 115200
3	PiControl CM4	USB 1 Port	2D QR Scanner	USB Cable	Molded USB-A	Barcode / Payment CDC
4	PiControl CM4	USB 2 Port	FHD Web Camera	USB Cable	Molded USB-A	1080p UVC Video Stream
5	PiControl CM4	USB 3 / 3.5mm	Audio Speaker	3.5mm Audio	Stereo Jack	Voice Chime Guidance
6	Driver X (HBS860H)	Terminal ALM-	PiControl CM4	Terminal DI0	0.5 mm² Red	X Drive Alarm Opto
7	Driver Y (HBS860H)	Terminal ALM-	PiControl CM4	Terminal DI1	0.5 mm² Red	Y Drive Alarm Opto
8	Driver X (HBS860H)	Terminal PEND-	PiControl CM4	Terminal DI2	0.5 mm² Amber	X In-Position Signal
9	Driver Y (HBS860H)	Terminal PEND-	PiControl CM4	Terminal DI3	0.5 mm² Amber	Y In-Position Signal
10	PiControl CM4	Terminal DO0	KM1 Contactor	Coil A1 (+24V)	1.0 mm² Orange	Safety Contactor Reset
11	PSU 2 (24V 5A)	Terminal +24V	PiControl CM4	24V IN (+)	1.0 mm² Orange	Aux 24V Logic Power
12	PSU 2 (24V 5A)	Terminal 0V	PiControl CM4	0V IN (-)	1.0 mm² Black	Field 0V Reference
13	PSU 2 (24V 5A)	Terminal +24V	IRiV IO Controller	Power +24V	1.5 mm² Orange	+24VDC Module Power
14	PSU 2 (24V 5A)	Terminal 0V	IRiV IO Controller	Power 0V	1.5 mm² Black	0VDC Module Return
15	IRiV IO Controller	Terminal 0V	IRiV IO Controller	Terminal S/S	Jumper Wire (Blk)	S/S tied to 0V (PNP)
16	IRiV IO Controller	Terminal DO0	Pilot Light Green	Lamp Anode (+)	0.75 mm² Green	Machine Ready Light
17	IRiV IO Controller	Terminal DO1	Pilot Light Yellow	Lamp Anode (+)	0.75 mm² Yellow	Carriage Moving Light
18	IRiV IO Controller	Terminal DO2	Red Light / Buzzer	Anode (+)	0.75 mm² Red	Alarm / Fault Indicator
19	IRiV IO Controller	Terminal DO3	Dispense Relay	Relay Coil A1	0.75 mm² Blue	Door Drop Solenoid
20	KM1 Contactor	Terminal 21-22	IRiV IO Controller	Terminal DI10	0.5 mm² Shielded	Fail-Safe NC E-Stop

PROJECT: NARIT SMART VENDING MACHINE | SECTION 3: FIELD SENSOR SYSTEM PIN-TO-PIN WIRING

SECTION: SECTION 3 | IEC 60204-1 / ISO 7200 STANDARDS

FIELD SENSOR WIRING SCHEMATIC
Pin-to-Pin Terminations of 3-Wire Proximity & Optical Sensors

SENSOR POWER DISTRIBUTION BUS (PSU 2)

+24VDC Bus Rail : Brown Wire (BN) on all 8 Sensors
0VDC Common Rail : Blue Wire (BU) on all 8 Sensors + S/S Terminal
PE Ground Bar : Shield Drain Wires Grounded at Cabinet PE

OPTICAL PHOTOELECTRIC SENSORS (OMRON E3Z-D81)

Completed Drop Sensor (DI8) :
Brown -> +24VDC | Blue -> 0VDC | Black -> IRiV IO Terminal DI8
Alarm / Pickup Door Sensor (DI9) :
Brown -> +24VDC | Blue -> 0VDC | Black -> IRiV IO Terminal DI9
Operating Mode : Dark-ON mode (Detects drop / door flap open)

INDUCTIVE PROXIMITY LIMIT SENSORS (PNP NC)

Standard Wiring for all 6 Limit Sensors (LJ12A3-4-Z/AX) :
Brown (BN) -> +24VDC Bus | Blue (BU) -> 0VDC Return Bus
Black (BK) -> Signal Output to IRiV IO Digital Inputs:
- Limit Min X (Home) -> Terminal DI0 (Channel 0)
- Limit Max X -> Terminal DI1 (Channel 1)
- Limit Min Y (Drop) -> Terminal DI2 (Channel 2)
- Limit Max Y (Top) -> Terminal DI3 (Channel 3)
- Limit Min Z (Home) -> Terminal DI4 (Channel 4)
- Limit Max Z (Push) -> Terminal DI5 (Channel 5)
Auxiliary NO Sensors :
- Z Home Datum Sensor -> Terminal DI6 | Drop Parking -> DI7

PIN-TO-PIN CONNECTION SCHEDULE (SECTION 3)
Sensor Leads -> Power Bus & IRiV IO Digital Input Terminals

NO	SENSOR NAME	LEAD / COLOR	TERMINATION POINT	PIN / TERM	WIRE SPEC / SHIELD	FUNCTION / ACTION
1	Completed Sensor	Brown Lead	Sensor Power Bus	Terminal +24V	3-Core 0.5 mm²	+24VDC Sensor Power
2	Completed Sensor	Blue Lead	Sensor Power Bus	Terminal 0V	3-Core 0.5 mm²	0VDC Power Return
3	Completed Sensor	Black Lead	IRiV IO Controller	Terminal DI8	3-Core Shielded	Drop Verification Beam
4	Alarm Sensor	Brown Lead	Sensor Power Bus	Terminal +24V	3-Core 0.5 mm²	+24VDC Sensor Power
5	Alarm Sensor	Blue Lead	Sensor Power Bus	Terminal 0V	3-Core 0.5 mm²	0VDC Power Return
6	Alarm Sensor	Black Lead	IRiV IO Controller	Terminal DI9	3-Core Shielded	Pickup Door Flap Open
7	Limit Min X Sensor	Brown Lead	Sensor Power Bus	Terminal +24V	3-Core 0.5 mm²	+24VDC Sensor Power
8	Limit Min X Sensor	Blue Lead	Sensor Power Bus	Terminal 0V	3-Core 0.5 mm²	0VDC Power Return
9	Limit Min X Sensor	Black Lead	IRiV IO Controller	Terminal DI0	3-Core Shielded	X Min Limit / Home (NC)
10	Limit Max X Sensor	Black Lead	IRiV IO Controller	Terminal DI1	3-Core Shielded	X Max Travel Limit (NC)
11	Limit Min Y Sensor	Black Lead	IRiV IO Controller	Terminal DI2	3-Core Shielded	Y Min (Drop Level NC)
12	Limit Max Y Sensor	Black Lead	IRiV IO Controller	Terminal DI3	3-Core Shielded	Y Max (Top Level NC)
13	Limit Min Z Sensor	Black Lead	IRiV IO Controller	Terminal DI4	3-Core Shielded	Z Min (Pusher Home NC)
14	Limit Max Z Sensor	Black Lead	IRiV IO Controller	Terminal DI5	3-Core Shielded	Z Max (Pusher Push NC)
15	Z Home Sensor	Black Lead	IRiV IO Controller	Terminal DI6	3-Core Shielded	Z Home Datum (NO)
16	Drop Park Sensor	Black Lead	IRiV IO Controller	Terminal DI7	3-Core Shielded	Carriage Alignment (NO)
17	All Sensor Cables	Drain Wire	Chassis Earth	TB-PE Bar	Braided Screen	Noise Shield Grounding
18	IRiV IO Controller	Power 0V	IRiV IO Controller	Terminal S/S	Jumper 1.0 mm²	S/S tied to 0V for PNP

MOTION CONTROL & PULSE INTERFACE SCHEMATIC

STM32 Hardware Timers, NMOS Open Drain Sink & Motor Terminals

STM32 NUCLEO-G491RE CO-PROCESSOR

Hardware Timer Output Compare Pulse Generation (10 - 50,000 Hz) :
X_STEP : PA8 (TIM1_CH1 / AF6) on Morpho CN10 pin 23 (Arduino D7)
X_DIR : PB0 (GPIO Output) on Morpho CN7 pin 34 (Arduino A3)
Y_STEP : PA9 (TIM1_CH2 / AF6) on Morpho CN10 pin 21 (Arduino D8)
Y_DIR : PB1 (GPIO Output) on Morpho CN10 pin 24
Z_STEP : PA5 (TIM2_CH1 / AF1) on Morpho CN10 pin 11 (Arduino D13)
Z_DIR : PB2 (GPIO Output) on Morpho CN10 pin 22 | GND : CN10 pin 20

6-CHANNEL LOGIC-LEVEL NMOS SINK BOARD

3.3V Logic Gate Drive -> 24V Low-Side Open-Drain Sink Circuits :
Gate Q1-Q6 : Driven by STM32 GPIOs via 100-Ohm series + 10k pulldown
Drain Q1-Q6 : Sinks Driver Terminals PUL- and DIR- to 0V
Source Q1-Q6 : Tied to MCU GND and Field 0VDC Reference
V-PULSE Bus (+24VDC) : Feeds all Driver PUL+ and DIR+ (Common Anode)

STEPPER MOTOR DRIVERS & ACTUATOR MOTORS

Driver X (HBS860H) & Motor X (86HBS85 NEMA 34) :
Power : +60VDC/0V | PUL/DIR from NMOS Ch1/Ch2 | Coils A/B | Encoder EA/EB
Driver Y (HBS860H) & Motor Y (86HBS85 NEMA 34 Elevator) :
Power : +60VDC/0V | PUL/DIR from NMOS Ch3/Ch4 | Coils A/B | Encoder EA/EB
Driver Z (DM542) & Motor Z (NEMA 17 V-Slot Pusher Actuator) :
Power : +24VDC/0V | PUL/DIR from NMOS Ch5/Ch6 | Coils A/B (Red,Blu,Blk,Grn)

PIN-TO-PIN CONNECTION SCHEDULE (SECTION 4)

STM32 Morpho Pins -> NMOS Board -> Motor Drivers -> Motor Coils & Encoders

NO	ORIGIN COMPONENT	PIN / TERM	TARGET COMPONENT	PIN / TERM	WIRE COLOR / GAUGE	SIGNAL / FUNCTION
1	STM32 G491RE	PA8 (CN10-23)	NMOS Board	Gate Q1 (In)	Ribbon Wire	X_STEP Pulse (TIM1)
2	STM32 G491RE	PB0 (CN7-34)	NMOS Board	Gate Q2 (In)	Ribbon Wire	X_DIR Direction
3	STM32 G491RE	PA9 (CN10-21)	NMOS Board	Gate Q3 (In)	Ribbon Wire	Y_STEP Pulse (TIM1)
4	STM32 G491RE	PB1 (CN10-24)	NMOS Board	Gate Q4 (In)	Ribbon Wire	Y_DIR Direction
5	STM32 G491RE	PA5 (CN10-11)	NMOS Board	Gate Q5 (In)	Ribbon Wire	Z_STEP Pulse (TIM2)
6	STM32 G491RE	PB2 (CN10-22)	NMOS Board	Gate Q6 (In)	Ribbon Wire	Z_DIR Direction
7	STM32 G491RE	GND (CN10-20)	NMOS Board	Source / 0V	Black 1.0 mm²	Digital Ground Return
8	PSU 2 (+24V)	Terminal +24V	Motor Drivers X,Y,Z	Term PUL+, DIR+	Orange 1.0 mm²	+24V Common Anode
9	NMOS Board	Drain Q1	Driver X (HBS860H)	Terminal PUL-	0.5 mm² Blue	X Step Opto Sink
10	NMOS Board	Drain Q2	Driver X (HBS860H)	Terminal DIR-	0.5 mm² Amber	X Dir Opto Sink
11	NMOS Board	Drain Q3	Driver Y (HBS860H)	Terminal PUL-	0.5 mm² Blue	Y Step Opto Sink
12	NMOS Board	Drain Q4	Driver Y (HBS860H)	Terminal DIR-	0.5 mm² Amber	Y Dir Opto Sink
13	NMOS Board	Drain Q5	Driver Z (DM542)	Terminal PUL-	0.5 mm² Blue	Z Step Opto Sink
14	NMOS Board	Drain Q6	Driver Z (DM542)	Terminal DIR-	0.5 mm² Amber	Z Dir Opto Sink
15	Driver X	Term A+ / A-	Motor X (86HBS85)	Phase A Leads	Red / Blue 1.5mm²	Motor Phase A Coils
16	Driver X	Term B+ / B-	Motor X (86HBS85)	Phase B Leads	Blk / Grn 1.5mm²	Motor Phase B Coils
17	Driver X	Term EA+/EA-	Motor X Encoder	Channel A Leads	Shielded Pair	Encoder Differential A
18	Driver X	Term EB+/EB-	Motor X Encoder	Channel B Leads	Shielded Pair	Encoder Differential B
19	Driver X	Term VCC/EGND	Motor X Encoder	Power Leads	Red / White 0.5mm²	+5V / 0V Encoder Power
20	Driver Y	Term A+ / A-	Motor Y (Elevator)	Phase A Leads	Red / Blue 1.5mm²	Motor Phase A Coils
21	Driver Y	Term B+ / B-	Motor Y (Elevator)	Phase B Leads	Blk / Grn 1.5mm²	Motor Phase B Coils
22	Driver Z	Term A+ / A-	Motor Z (Pusher)	Phase A Leads	Red / Blue 0.75mm²	Motor Phase A Coils
23	Driver Z	Term B+ / B-	Motor Z (Pusher)	Phase B Leads	Blk / Grn 0.75mm²	Motor Phase B Coils